A CASE-STUDY BASED REPORT

ON

Parking Space Availability Prediction Using Random Forest and Time-Series Models

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ABSTRACT

Parking Space Availability Prediction Using Random Forest and Time-Series Models is a smart parking management system designed to predict the availability of parking spaces at a particular location and time. Finding a parking space has become a major challenge in cities due to the increasing number of vehicles. Drivers often spend a lot of time searching for empty parking spaces, which leads to traffic congestion, fuel wastage, air pollution, and unnecessary delays. This project aims to solve this problem by using machine learning and forecasting techniques to provide users with accurate information about parking availability before they reach their destination. The system uses historical parking data, including date, time, day of the week, occupancy level, holidays, and other related factors, to make reliable predictions.

The project uses the Random Forest algorithm to classify whether a parking space is available or occupied based on the collected data. It also uses Time-Series models to study past parking patterns and predict future parking demand during different times of the day. The predicted results are displayed through a simple web application or dashboard, allowing users to check parking availability in advance and plan their journey more efficiently. This system helps reduce traffic congestion, save fuel and travel time, improve the utilization of parking spaces, and support the development of smart city transportation systems. It is a simple, efficient, and user-friendly solution that benefits both drivers and parking management authorities.

1. INTRODUCTION

1.1 Background

Urban parking demand varies by location, hour and day. Drivers who cannot quickly identify a suitable space may circulate through busy streets, adding to congestion, fuel consumption and emissions. Historical occupancy records can reveal recurring demand patterns and help estimate when a car park is likely to be busy.

1.2 Motivation for the Project

The project aims to make parking information easier to interpret by turning historical occupancy observations into availability estimates and short-term forecasts. It combines classification, time-series analysis and a web interface so users can compare car parks and inspect expected demand patterns.

1.3 Overview of Work Carried Out

The supplied project contains processed parking data, a saved Random Forest model, feature and car-park metadata, time-series forecast artefacts, evaluation reports, and a Flask application. The implementation engineers calendar features, evaluates the classifier using a chronological split, calculates historical occupancy quantiles, displays three occupancy-status tiers, and records prediction history in SQLite.

1.4 Scope and Responsible Use

The system is a historical-data-based decision-support prototype. Its outputs are estimates, not guarantees that a physical space will be free at arrival. Users and parking operators should treat predictions as guidance and validate them against live sensors or operational systems before relying on them for critical decisions.

2. PROBLEM STATEMENT

2.1 Case-Study Scenario

A driver planning a visit to a busy area needs to choose among multiple car parks. Availability can change over the day, and a car park that is usually quiet may become crowded during a recurring peak period. The system uses historical patterns to estimate occupancy for a selected car park and time.

2.2 Central Case-Study Question

How can historical parking occupancy and time-related patterns be used to estimate parking availability and forecast near-term occupancy for individual car parks?

2.3 Formal Problem Definition

Given a car-park identifier, its capacity and a timestamp, the system derives calendar features and estimates the likely occupancy/availability category. Separately, the time-series components produce future occupancy estimates at 30-minute intervals. The operational status thresholds are Available below 70% occupancy, Near Full from 70% to below 90%, and Full at or above 90%.

2.4 Objectives

- Explore historical occupancy patterns by time, day and car park.
- Train and evaluate a Random Forest classifier using engineered time and car-park features.
- Use time-series model artefacts to represent expected short-term occupancy changes.
- Provide a dashboard with lot-level status, forecast visualisation and prediction history.
- Identify limitations caused by class imbalance and the absence of live occupancy inputs.

2.5 Assumptions and Limitations

The historical dataset represents Birmingham car parks during a limited period in late 2016. Demand patterns may have changed since then. The model features do not include a live or recent occupancy measurement, and the target distribution is imbalanced. The displayed estimate is therefore based on historical occupancy quantiles rather than treating the raw Random Forest probability as a reliable live availability probability.

3. DATASET AND EXPLORATORY DATA ANALYSIS

3.1 About the Dataset

The project README and report identify the source as the Birmingham parking dataset. The processed data contains 35,717 readings for 30 car parks from 2016-10-04 through 2016-12-19, recorded at approximately 30-minute intervals. The supplied report states that the dataset has no missing values.

Field / derived feature	Description
SystemCodeNumber	Identifier for a parking lot.
Capacity	Total capacity recorded for the car park.
Occupancy	Observed occupied spaces.
LastUpdated	Timestamp associated with the reading.
OccupancyRate	Occupancy divided by capacity.

Hour, Minute, DayOfWeek	Time components derived from the timestamp.
IsWeekend, IsHoliday	Calendar indicator features.
TimeSlot, Week, Month	Additional time-related features.
LotID	Integer-encoded car-park identifier.
IsAvailable	Binary label: 1 when occupancy rate is below 0.90; otherwise 0.

3.2 Purpose of EDA

Exploratory data analysis was used to inspect class balance, occupancy distribution and demand variation by hour, day of week and individual car park. These patterns help explain the forecasting task and highlight where availability estimates are likely to be most useful.

3.3 Preparation Steps Actually Performed

- Use the processed parking CSV supplied with the project.
- Derive calendar features from timestamps, including hour, minute, weekday, weekend, holiday, time slot, week and month.
- Calculate OccupancyRate from Occupancy and Capacity.
- Create the IsAvailable label using a 90% occupancy threshold.
- Use a chronological 80% training / 20% testing split for classifier evaluation, keeping later observations for testing.

3.4 EDA Exploration and Findings



Figure 1. Class balance for the derived availability label.

The project report describes the label distribution as approximately 89.1% Available and 10.9% Full at the 90% threshold. This imbalance is important when interpreting accuracy and class-specific performance.

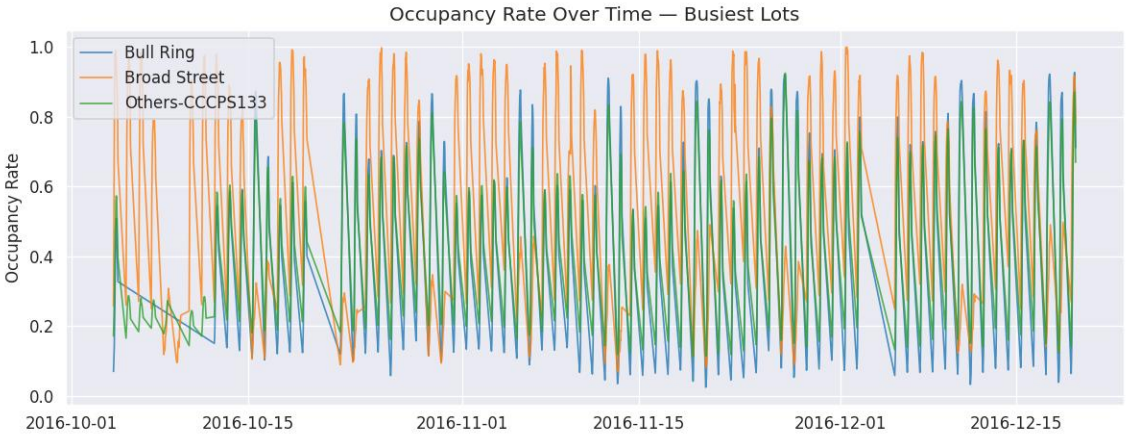


Figure 2. Occupancy trends over time for selected busy car parks.

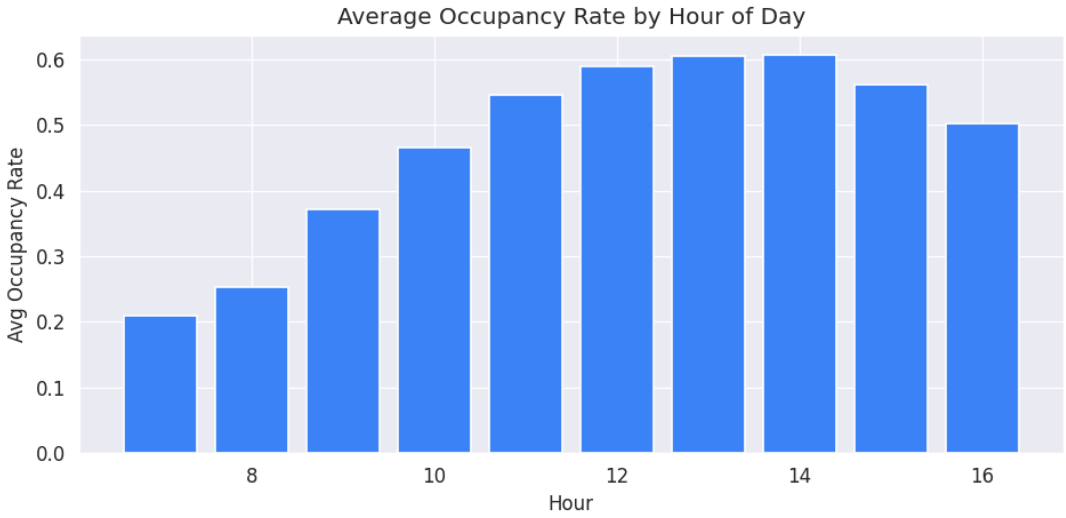


Figure 3. Average occupancy by hour of day.

The hourly analysis identifies a daytime/midday rise in occupancy, showing that availability is not constant throughout the day.

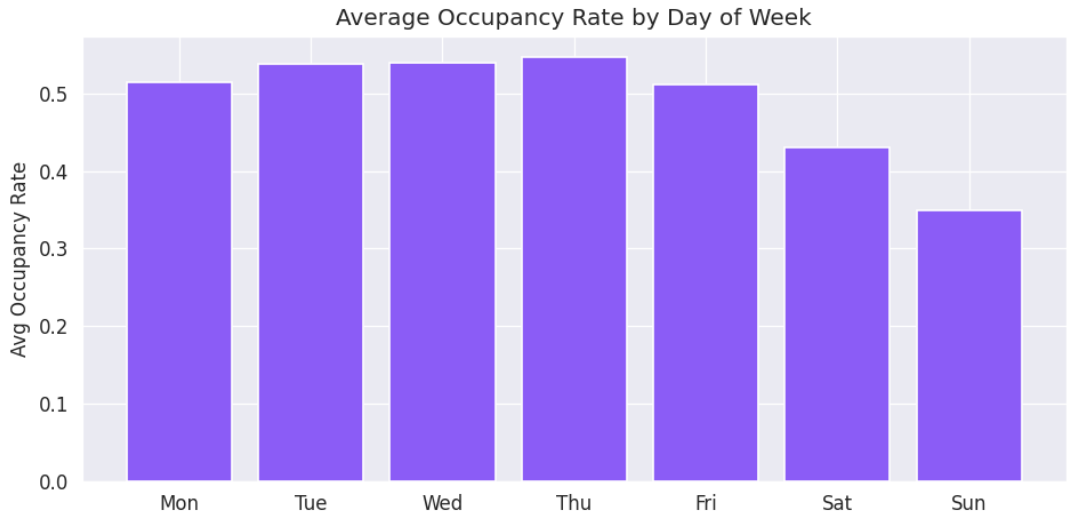


Figure 4. Average occupancy by day of week.

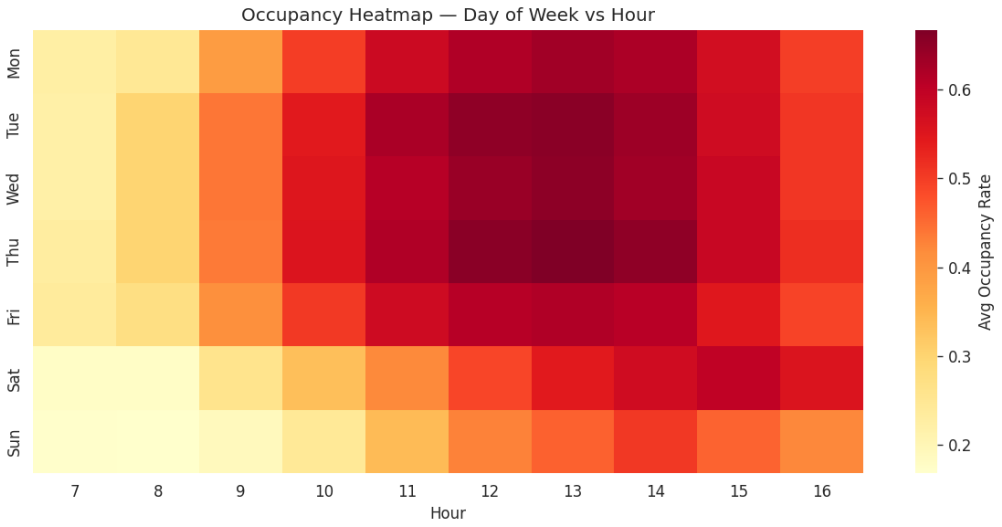


Figure 5. Average occupancy by weekday and hour.

The day-and-hour analysis highlights higher weekday demand around midday, particularly in the 11:00–14:00 period, as described in the supplied project report.

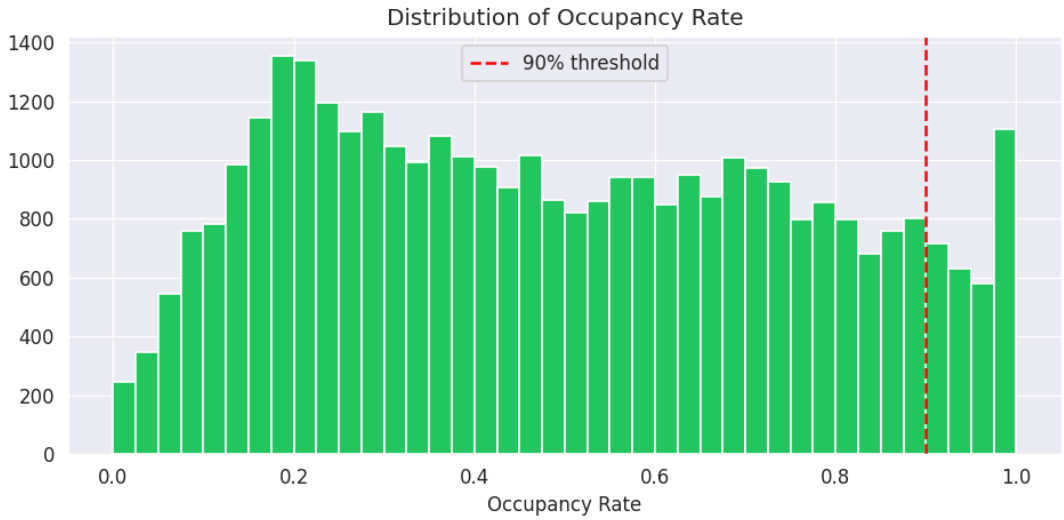


Figure 6. Distribution of occupancy rate relative to the 90% threshold.

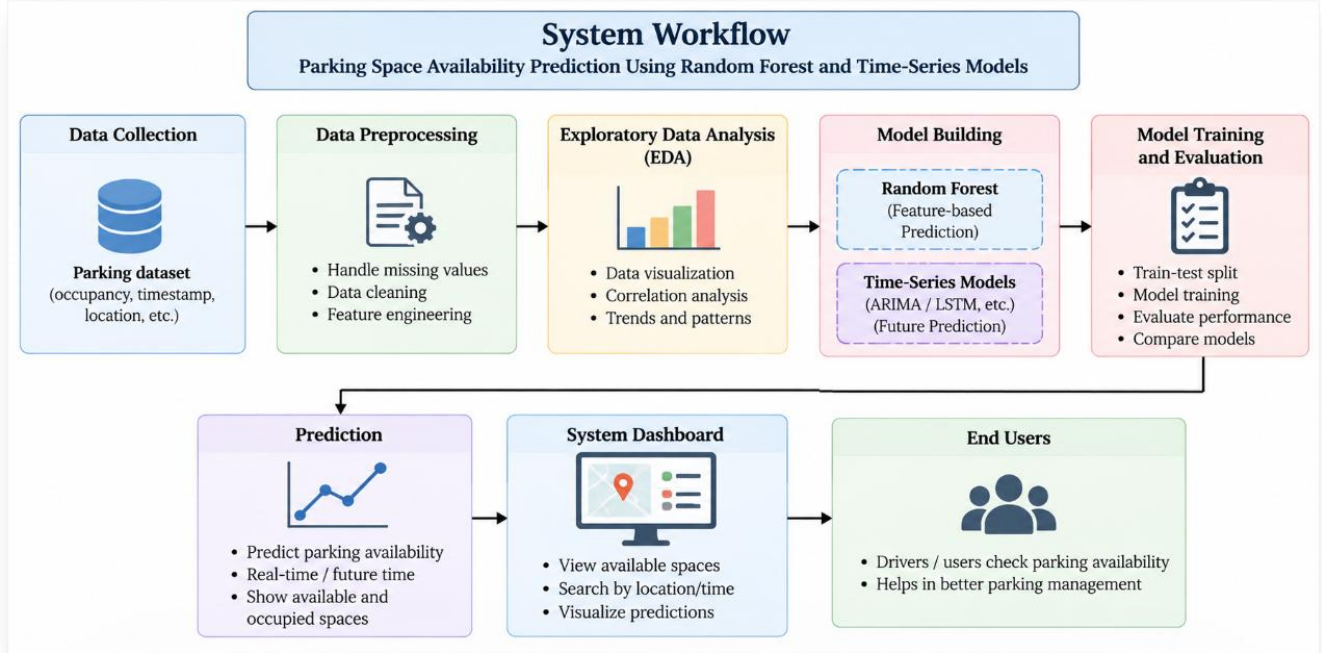
3.5 Summary of EDA Findings

Occupancy varies by hour, day and car park. The derived target is imbalanced toward Available. These findings motivate time-based features and the use of historical lot-specific patterns, while also cautioning against interpreting accuracy alone as evidence of balanced performance.

4. ALGORITHM SELECTION AND IMPLEMENTATION

4.1 System Workflow

The pipeline loads processed parking data and metadata, builds calendar and lot features, applies the trained Random Forest model for classification evaluation, and uses historical occupancy quantiles to generate the dashboard estimate. Time-series forecast files and a saved Holt–Winters model support forecast visualisation. The Flask application exposes the results through a browser interface and stores prediction history in SQLite.



4.2 Random Forest Classifier

The supplied report identifies a Random Forest classifier with 200 trees, maximum depth 15 and balanced class weights. The model uses LotID, Hour, Minute, DayOfWeek, IsWeekend, IsHoliday, TimeSlot, Week, Month and Capacity. A Random Forest combines predictions from multiple decision trees, which can capture nonlinear relationships among these features.

4.3 Time-Series Forecasting

Time-series forecasting is used to predict future parking space availability by analyzing historical parking occupancy data and identifying patterns over time. The project uses two forecasting methods: Holt–Winters Exponential Smoothing and SARIMA.

Holt–Winters Exponential Smoothing considers the level, trend, and seasonality in the data to forecast future parking occupancy. The project uses an additive trend and seasonality configuration with a seasonal period of 48 observations, representing a daily cycle when data is recorded every 30 minutes.

SARIMA (Seasonal Autoregressive Integrated Moving Average) is another time-series model used to capture patterns and seasonal changes in parking occupancy. The project documents a SARIMA(1,0,1)(1,0,1,48) configuration.

These models help identify recurring parking patterns and estimate future occupancy levels. The forecasts can support drivers in planning their parking and help parking managers improve space utilization.

The project includes saved forecasting artifacts, including a trained Holt–Winters model and time-series forecast outputs. These artifacts support the analysis of predicted parking availability over time.

4.4 Availability Estimate and Status Rules

The prediction module retrieves the historical 90th-percentile occupancy rate for the selected car park, weekday and hour. If that exact combination is unavailable, it falls back to lot-and-hour, then lot-level, and finally global historical quantiles. Estimated availability is calculated as 100% minus the estimated occupancy rate. The dashboard labels occupancy below 70% as Available, from 70% to below 90% as Near Full, and 90% or higher as Full. The raw Random Forest probability is retained for transparency/debugging but is not used as the primary displayed status.

4.5 Web Application and Storage

The application uses Python and Flask for the web service, HTML templates for the interface, and SQLite for prediction history. The dashboard includes a prediction panel, a status overview of the 30 lots, forecast charts, historical prediction records and model metrics. The README provides the local run command: install the requirements, change to the app directory, and run python app.py.

5. RESULTS AND DISCUSSION

5.1 Random Forest Evaluation

Metric	Reported result
Accuracy	0.9178 (91.78%)
ROC-AUC	0.9652
Average precision	0.9943
Test records	7,144

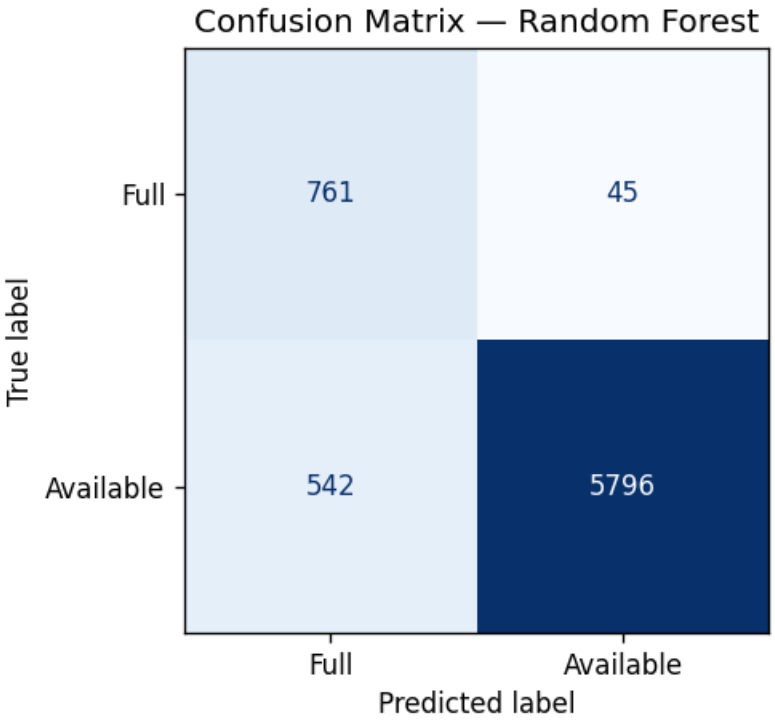


Figure 7. Confusion matrix on the chronological test set.

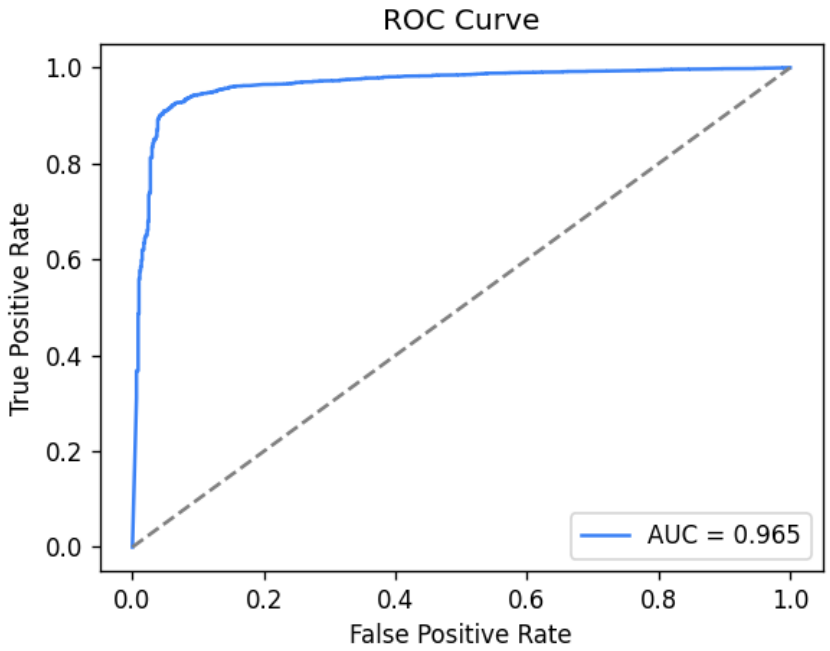


Figure 8. ROC curve; reported ROC-AUC = 0.9652.

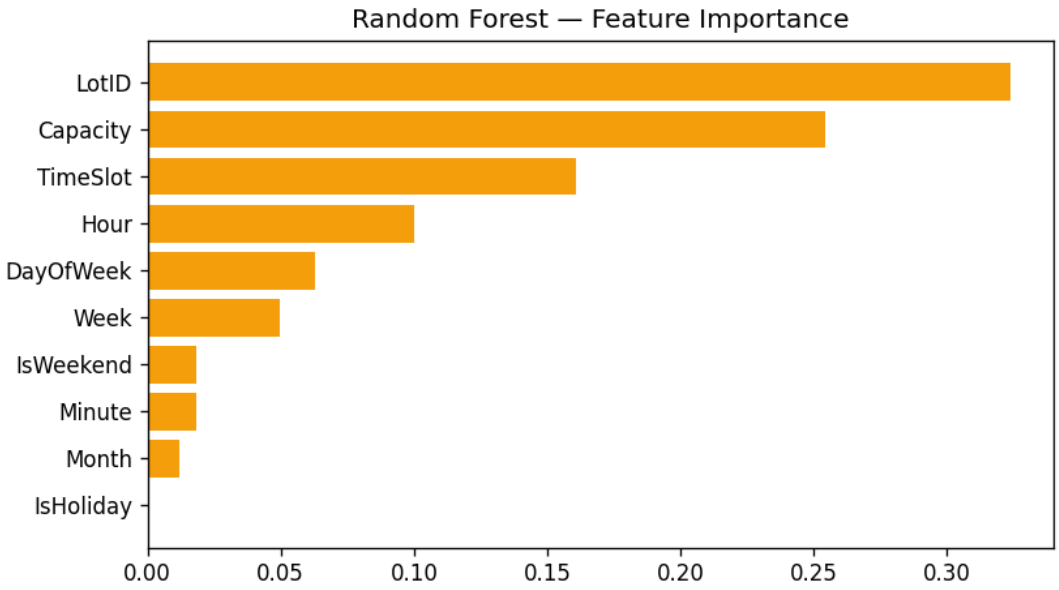


Figure 9. Random Forest feature importance.

The supplied evaluation report gives Full-class precision of 0.58, recall of 0.94 and F1-score of 0.72. For Available, precision is 0.99, recall is 0.91 and F1-score is 0.95. These class-specific results show why the aggregate accuracy must be considered alongside the minority-class metrics. The report notes that LotID and Capacity dominate the feature importance ranking.

5.2 Interpretation and Deployment Consideration

The project documentation records that raw Random Forest probabilities were greater than 0.99 for Available for nearly every tested lot/time combination. The stated cause is that the feature set lacks actual or recent occupancy and approximately 89% of the labels are Available. The deployed prediction function therefore bases the displayed

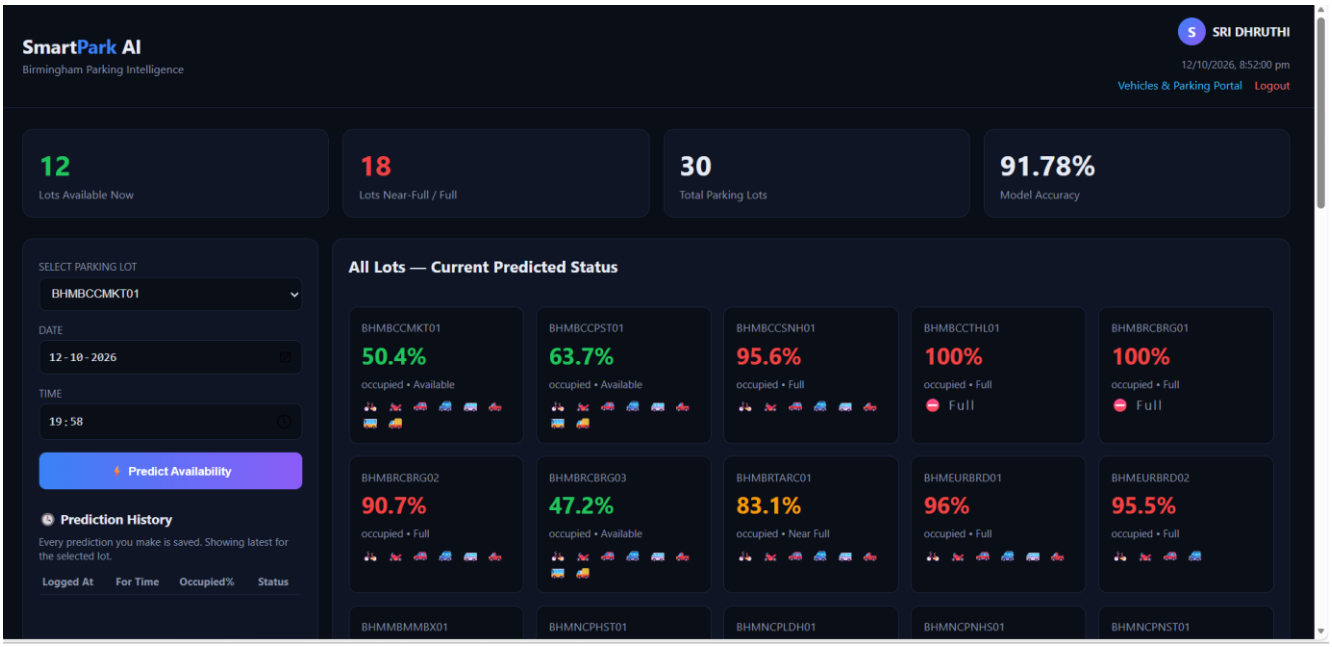
estimate on historical occupancy quantiles. This is an explicit engineering adjustment and should not be interpreted as proof that the classifier itself predicts live occupancy accurately.

5.3 Time-Series Forecasts

The project report states that Holt–Winters and SARIMA forecasts were generated for each lot for 24–48 hours ahead and generally tracked expected overnight decreases and daytime increases. The provided application artefacts include time-series forecasts and a Holt–Winters model. Forecast reliability will depend on how closely current conditions resemble the historical period.

6. SYSTEM DASHBOARD

The SmartPark AI web application is designed to let users select a car park and date/time, inspect the estimated availability percentage and status, view a grid covering the 30 lots, review near-term forecast charts, and retrieve past predictions. Prediction history is stored in an SQLite database. The supplied ZIP includes Flask routes and HTML templates for the interface; this report describes the supplied implementation rather than claiming live sensor integration.



6.2 Key Features and Functionalities

The SmartPark AI dashboard provides a user-friendly interface for monitoring parking space availability and viewing predictions. It combines machine learning and time-series forecasting to help users understand parking occupancy patterns.

The key features of the dashboard include:

- **Forecast:** Displays historical parking occupancy and predicted occupancy for future periods.
- **Hourly Analysis:** Provides parking information at hourly intervals.
- **Weekly Analysis:** Helps analyze parking patterns over longer periods.
- **Model Information:** Provides information about the prediction models.
- **Historical Occupancy:** Displays past occupancy rates for the selected car park.

- **90% Occupancy Threshold:** Highlights the occupancy limit used to identify nearly full parking areas.
- **Time-Series Forecasting:** Uses historical patterns to estimate future parking occupancy.
- **Visual Graphs:** Presents occupancy trends and predictions through line charts.

These features help users understand parking conditions and support better parking management decisions.



6.3 Parking Availability Predictions and Visualizations

The system predicts parking availability using a Random Forest classifier and analyzes future occupancy patterns using time-series forecasting methods.

The Random Forest model classifies parking availability based on a predefined occupancy threshold of 90%. Parking occupancy below this threshold is classified as available, while occupancy at or above the threshold is classified as near-full or full.

The dashboard presents prediction results through visualizations that make parking patterns easier to understand.

The main visualizations include:

- **Parking Occupancy Graph:** Displays historical parking occupancy over time.
- **Occupancy Threshold Line:** Shows the 90% occupancy threshold to identify crowded parking areas.
- **Time-Series Forecast Graph:** Presents forecasted parking occupancy for future time periods.
- **Parking Status Display:** Shows the availability status of individual car parks.
- **Prediction Summary:** Presents model performance information, including the reported Random Forest accuracy of 91.78%.

These visualizations help users identify parking trends, understand occupancy changes, and make informed decisions about parking.

7. RESPONSIBLE USE, LIMITATIONS AND FUTURE WORK

7.1 Limitations

- The dataset covers a limited historical period in 2016 and may not reflect current parking demand.
- The classifier target is imbalanced, so accuracy alone can obscure weaker performance on the Full class.

- The classifier features do not include a live occupancy sensor reading.
- Historical quantile estimates represent typical busy-period patterns and cannot guarantee a free space.
- The report documents forecast methods and artefacts, but does not provide a complete independent comparison of all forecasting metrics.

7.2 Future Enhancements

- Integrate reliable live sensor or operator occupancy feeds.
- Retrain and validate models on a more recent, longer dataset.
- Add temporal back testing and compare forecasting errors across car parks.
- Calibrate status estimates and evaluate performance separately for each occupancy tier.
- Add monitoring for data drift, missing feeds and stale forecasts.

8. CONCLUSION

This project demonstrates a practical parking availability analysis system using historical occupancy data, a Random Forest classifier, time-series forecasting artefacts and a Flask dashboard. The dataset contains 35,717 readings across 30 Birmingham car parks, and the saved classifier achieved 91.78% accuracy and 0.9652 ROC-AUC on the reported chronological test split. The project also identifies an important limitation: time and car-park feature alone led the raw classifier to strongly favour the majority Available class. The deployed estimate therefore uses historical occupancy quantiles to provide more differentiated lot-and-time estimates. Future work should prioritise current data, live occupancy integration and systematic forecast validation before operational deployment.

REFERENCES

- Birmingham parking occupancy dataset, as identified in the supplied SmartPark AI project documentation (source attribution in the project report: Kaggle / UCI).
- Scikit-learn documentation, Random Forest and classification evaluation tools.
- Statsmodels documentation, time-series forecasting and exponential smoothing.
- Flask documentation, web application framework.
- SmartPark AI source files and project documentation supplied in the Team 17 project ZIP.

TEAM SIGNATURES